

Notes: Azmat and Ferrer (JPE, 2017)

Gender Gaps in Performance: Evidence from Young Lawyers

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1 Big picture

- **Question.** Do gender gaps in career outcomes among high-skilled professionals partly reflect gender gaps in actual workplace performance rather than only discrimination or differences in hours worked?
- **Setting.** The paper studies young lawyers in the United States using the *After the JD* longitudinal survey. The core performance analysis uses the 2007 wave, which contains profession-standard performance measures, and the promotion analysis links those lawyers to partnership outcomes in 2012.
- **Core challenge.** A gender gap in earnings or promotion does not by itself tell us whether women and men are being rewarded differently for the same performance, whether they perform differently, or whether family structure, career concerns, or workplace frictions affect performance itself.
- **Main conceptual contribution.** The paper studies *direct* measures of performance in a high-skill profession that evaluates lawyers using transparent and homogeneous metrics: annual hours billed and new client revenue. This is a major step beyond treating hours worked as a proxy for productivity.
- **Main hypotheses.** The paper studies three central explanations for the performance gap:
 - employer discrimination and related workplace frictions,
 - the presence of children, especially preschool-aged children,
 - differences in aspirations to become a partner, interpreted as career concerns.

It also studies area of specialization, overbilling, networking, and weekend work.

- **Main empirical design.** The authors estimate regressions of performance, earnings, and later partnership on a female indicator plus rich controls for demographics, firm characteristics, educational proxies for ability, and region fixed effects. They then add interaction terms and alternative mechanisms one by one.
- **Main baseline result.** Male lawyers bill about 153 more hours per year than female lawyers, roughly a 10% gap, and they bring in about \$30,000 more annual new client revenue. These gaps remain large after adding extensive controls.
- **Key mechanism result.** Children under age 4 reduce female lawyers' billed hours substantially but do not affect male lawyers' billed hours in the same way. Differences in aspirations to become partner are especially important for explaining the gender gap in new client revenue.
- **Key credibility result.** The paper finds no meaningful gender gap in target billable hours, little explanatory power from direct discrimination measures, and no evidence that women time fertility around periods of unusually low prior performance.
- **Main career-outcome result.** Performance explains a large share of the remaining gender gap in annual earnings and a sizable share of the partnership gap. Once performance is accounted for, the partnership gap is no longer statistically significant.
- **Economic takeaway.** Even in a very elite profession with highly educated men and women, gender gaps in family constraints and career aspirations translate into meaningful performance gaps, and those performance gaps matter a great deal for pay and promotion.

2 Data and measurement (key objects)

2.1 Why the legal profession is a useful laboratory

- The legal profession uses standardized, transparent performance metrics across firms and legal fields.
- The two core measures are **hours billed** and **new client revenue**. Law firms use them not only for bonus pay but also for promotion evaluations.
- This makes the profession especially useful for separating performance gaps from outcome gaps.
- The setting is also informative because the cohort is highly skilled and comes from a period with near gender parity in law school admissions.

Interpretation. The paper is not inferring productivity indirectly. It studies the variables that law firms themselves use to evaluate lawyers.

2.2 Data source and sample construction

- The data come from *After the JD*, a nationally representative longitudinal survey of U.S. lawyers first admitted to the bar in 2000.
- Lawyers were surveyed in 2002 and again in 2007; later promotion outcomes are observed in 2012.
- The 2007 wave is central because it includes hours billed, new client revenue, and aspirations to become partner.
- The paper focuses on lawyers who bill hours. More than 93% of these respondents work in law firms, with the remainder mainly in solo practices.

Interpretation. The core sample is closely aligned with the institutional setting where billable hours and client origination matter most.

2.3 Hours billed

Let

HB_i = annual number of hours billed by lawyer i in 2006.

- This is the first main performance measure.
- It is not the same as hours worked. Lawyers often work more than they can bill because some tasks are not client-specific.
- The paper also observes **target hours to bill**, the firm's billable-hours requirement for the lawyer's position.

Interpretation. Hours billed are the profession's direct revenue-based productivity measure. They capture what the firm can actually charge, not simply time spent at work.

2.4 New client revenue

Let

NCR_i = new client revenue personally brought to the firm by lawyer i in 2006.

- This is the second main performance measure.
- It captures “rainmaking” or origination: the ability to attract new clients and generate business for the firm.
- It also contains an important long-run quality and reputation component because referrals and future visibility matter for generating new business.

Interpretation. Relative to hours billed, new client revenue is the more forward-looking performance measure. It is especially relevant for partnership because partners must help support the firm’s client base.

2.5 Earnings, partnership, and other outcomes

- **Annual earnings** are total earnings, defined as straight salary plus bonus.
- **Partnership** is an indicator for whether the lawyer is an equity partner or shareholder in 2012.
- The paper also studies:
 - target hours,
 - hours worked,
 - performance per hour worked,
 - and selection into jobs that require billing hours.

2.6 Controls and ability proxies

The main control vectors include:

- demographics: age, marriage, number of children, child under age 4, ethnicity;
- job characteristics: tenure, full-time status, firm type, firm size;
- education proxies: law school ranking, undergraduate ranking, judicial clerkship, moot court, journal participation;
- region fixed effects.

Interpretation. The paper tries hard to avoid attributing to gender what could instead reflect skill, job type, or family structure.

2.7 Descriptive facts

Table 1 gives the key sample facts:

- Male lawyers earn \$150,667 on average, compared with \$132,685 for female lawyers.
- Male lawyers bill 1,826 annual hours on average; female lawyers bill 1,677.
- Male lawyers generate \$53,346 in new client revenue on average; female lawyers generate \$23,349.
- Male lawyers report target hours of 1,827, compared with 1,759 for women.
- Men work 54.09 hours per week on average; women work 48.83.
- Women are somewhat younger, less likely to be married, and have fewer children, but they do not differ significantly in tenure or law-firm employment rates.

Interpretation. The raw gaps are already visible in the basic summary statistics, especially for billed hours and client origination.

3 Models and empirical specifications (all equations + interpretation)

3.1 Economic mechanism and hypotheses

The paper studies a reduced-form question: why do women and men in the same profession differ in performance? The authors emphasize three main channels:

- **Workplace discrimination or allocation frictions:** women may receive fewer assignments, more write-downs, or less favorable mentoring.
- **Child rearing:** young children may affect women's and men's performance differently.
- **Career concerns:** stronger aspirations to become partner may raise effort and client-development activity.

Additional channels include specialization, overbilling, networking, and weekend work.

Interpretation. The paper is not built around a structural model of effort. Instead, it uses the institutional features of law firms to compare a set of plausible mechanisms.

3.2 Baseline performance regression

For lawyer i , let

$$Perf_i \in \left\{ \frac{HB_i}{1000}, \frac{NCR_i}{100000} \right\}.$$

The baseline specification is

$$Perf_i = \alpha + \beta Female_i + X_i' \gamma + F_i' \delta + E_i' \theta + \lambda_{r(i)} + \varepsilon_i, \quad (1)$$

where:

- $Female_i$ is an indicator for women,
- X_i contains demographic controls,

- F_i contains firm controls,
- E_i contains educational proxies for ability,
- $\lambda_{r(i)}$ are region fixed effects.

Interpretation. The coefficient β is the conditional gender gap in performance after comparing lawyers with similar observed background and firm characteristics.

3.3 Target-hours specification

To study whether firms simply expect less from women, the paper estimates

$$Target_i = \alpha + \beta Female_i + X_i' \gamma + F_i' \delta + E_i' \theta + \lambda_{r(i)} + u_i, \quad (2)$$

where $Target_i$ is annual target billable hours, and in a related specification the dependent variable is an indicator for reporting zero target hours.

Interpretation. If the billed-hours gap merely reflected lower firm expectations or different job assignment rules, then a sizable negative coefficient on $Female_i$ should appear here. The paper does not find that pattern.

3.4 Performance relative to hours worked

The authors compare output to labor supply using:

$$\frac{HB_i}{HW_i}, \quad \frac{NCR_i}{HW_i},$$

where HW_i is annual hours worked.

They also define aggregate performance in hour-equivalent units:

$$AggPerf_i = HB_i + \frac{NCR_i}{200}, \quad (3)$$

using \$200 per hour as a conversion rate from client revenue to billable-hours equivalent.

Interpretation. These checks ask whether the gender gap is only a labor-supply gap or whether performance per hour worked also differs. The answer is that hours worked alone is not enough.

3.5 Child-rearing interaction specification

The main family specification is

$$\begin{aligned} Perf_i = & \alpha + \beta Female_i + \sum_k \lambda_k Child_{ik} + \phi YoungChild_i + \sum_k \rho_k (Female_i \times Child_{ik}) \\ & + \tau (Female_i \times YoungChild_i) + X_i' \gamma + F_i' \delta + E_i' \theta + \lambda_{r(i)} + \varepsilon_i. \end{aligned} \quad (4)$$

where the child indicators distinguish one child, two children, three or more children, and whether the lawyer has a child under age 4.

Interpretation. The coefficient τ is the key term. It captures whether young children affect female lawyers differently from male lawyers.

3.6 Career-aspiration specification

Aspirations are measured on a 1–10 scale, where higher values indicate stronger aspirations to become an equity partner. The paper estimates

$$Perf_i = \alpha + \beta Female_i + \eta Asp_i + \rho(Female_i \times Asp_i) + X_i'\gamma + F_i'\delta + E_i'\theta + \lambda_{r(i)} + \varepsilon_i. \quad (5)$$

Interpretation.

- $\eta > 0$ means higher aspirations are associated with higher performance.
- If $\rho \approx 0$, then aspirations affect men and women similarly; what matters is that women and men are distributed differently across aspiration levels.

3.7 Predicted-aspirations specification

To reduce concern that current aspirations merely reflect current firm feedback, the paper predicts 2007 aspirations using variables measured earlier:

$$Asp_i^{2007} = \pi_0 + \pi_1 SatLaw_i^{2002} + \pi_2 Stay_i^{2002} + Z_i'\pi + \nu_i, \quad (6)$$

where $SatLaw_i^{2002}$ is satisfaction with the decision to become a lawyer and $Stay_i^{2002}$ is the desired duration with the current employer.

The predicted component is then used in the performance regressions.

Interpretation. This is not a formal IV design with a sharp exclusion restriction. It is a way to isolate the more predetermined part of aspirations, before most career feedback in the legal profession had accumulated.

3.8 Earnings specification

The earnings equation is

$$\ln(Earnings_i) = \alpha + \beta Female_i + \psi HW_i + \kappa HB_i + \mu NCR_i + X_i'\gamma + F_i'\delta + E_i'\theta + \lambda_{r(i)} + \varepsilon_i. \quad (7)$$

The paper also adds:

- areas of law,
- squared and cubic terms in HB_i and NCR_i ,
- and specifications that replace HB_i with hours worked.

Interpretation. This is the central comparison between labor supply and actual performance. If performance is what law firms reward, then controlling for billed hours and new client revenue should shrink the remaining gender earnings gap more than controlling only for hours worked.

3.9 Promotion specification

For 2012 partnership outcomes, the paper estimates a linear probability model:

$$\begin{aligned} Partner_i^{2012} = & \alpha + \beta Female_i + \psi HW_i^{2007} + \kappa HB_i^{2007} + \mu NCR_i^{2007} + \eta Asp_i^{2007} \\ & + X_i' \gamma + F_i' \delta + E_i' \theta + \lambda_{r(i)} + \varepsilon_i. \end{aligned} \quad (8)$$

Interpretation. This links earlier measured performance to later career advancement. The question is whether the partnership gap survives after controlling for the specific metrics the profession values.

4 Identification and instruments (what makes the estimates credible)

4.1 What identifies the baseline estimates

The paper does *not* use an instrument or experimental variation. Identification is based on:

- rich observed controls,
- educational proxies for ability,
- region fixed effects,
- and a sequence of mechanism-specific tests.

Interpretation. The paper's strength is not quasi-experimental identification in the narrow sense. Its strength is careful use of profession-specific measures plus many direct checks against obvious competing explanations.

4.2 Why simple firm-expectation stories are limited

A basic concern is that women might bill less simply because firms expect them to bill less. The paper addresses this by studying target billable hours and the probability of reporting zero target hours.

- In the target-hours regressions, there is no statistically meaningful gender gap once the same controls are added.
- There is also no gender difference in the probability of reporting zero target hours.

Interpretation. The billed-hours gap is not just a mirror of lower formal expectations.

4.3 Discrimination channels examined directly

The paper studies several direct channels through which discrimination could affect measured performance:

- not receiving enough assignments,
- partners discounting or writing down hours,
- the gender and seniority of mentors,
- tasks involving contact with clients,
- and self-reported discrimination experiences.

Main findings:

- being constrained by assignments matters for billed hours in levels, but it does not explain the female coefficient;
- partner write-downs do not explain the gap;
- having a senior male mentor does not eliminate the performance gap;
- client-facing tasks help explain new client revenue in levels, but not the gender gap;
- direct discrimination measures do not have strong explanatory power for the performance gap.

Interpretation. The paper does not claim discrimination is absent in the profession. It shows that the direct channels it can measure do not account for much of the observed performance gap.

4.4 Child rearing and fertility selection

One of the strongest results is that women with children under age 4 bill substantially fewer hours, while men with young children do not experience the same decline.

To address selection concerns, the paper checks:

- **cross-sectional selection:** whether women with children are drawn from the low end of the predicted performance distribution;
- **timing selection:** whether men or women experience a performance “dip” before first birth.

The authors find no evidence that women with children were already negatively selected on predicted performance, and no evidence that births are timed after unusually weak prior performance.

Interpretation. The family result does not look like a simple artifact of women with lower expected performance choosing motherhood or timing births around weak years.

4.5 Aspirations and reverse causality

A central concern is that aspirations may be endogenous: perhaps people say they want to become partner because they are already doing well.

The paper addresses this in three ways:

- it adds many contemporaneous workplace and demographic controls;
- it uses 2002 variables to predict 2007 aspirations;
- it studies later partnership outcomes directly.

Interpretation. The aspiration evidence is not perfectly causal, but it is considerably stronger than a simple contemporaneous correlation between ambition and success.

4.6 Alternative behaviors that do not explain much

The paper also studies:

- overbilling,
- networking time,
- weekend work,
- and area of specialization.

These variables often matter for *levels* of performance. For example, networking and weekend work are positively associated with client revenue and billed hours. But they explain very little of the gender gap itself.

Interpretation. This distinction matters: many factors raise performance, but only a few factors explain why women's performance differs from men's.

4.7 Selection into billable-hours jobs

The authors also ask whether the female billed-hours gap is driven by who selects into jobs that require billing hours.

- Women are slightly less likely to enter such jobs.
- But the evidence suggests it is not lower-ability women who are selecting into billable-hours jobs.

Interpretation. Selection into the billable-hours sample does not appear to be the main reason women bill fewer hours within that sample.

4.8 What the paper can and cannot establish

- The paper is strongest on documenting that direct performance gaps exist and that they matter for earnings and promotion.
- It is also strong on showing that children and aspirations are empirically important mechanisms.
- It is weaker on fully separating internal preferences, social norms, anticipated discrimination, and realized discrimination as deep causes of aspiration differences.
- Because the performance data are self-reported, some measurement error is possible, although the authors note that external sources look broadly similar.

Interpretation. This is a persuasive empirical paper, but not a definitive causal decomposition of all underlying social mechanisms.

4.9 Relation to the literature

The paper contributes to three literatures at once:

- gender gaps in earnings,
- the glass ceiling and promotion gaps among elite professionals,
- and the economics of performance pay and career concerns.

Its distinctive contribution is to show that an omitted variable — actual profession-specific performance — explains an important part of what would otherwise look like an unexplained earnings or promotion gap.

5 Tables and figures

5.1 Figure 1: evolution of lawyers' gender gap in earnings

Read:

- Median weekly earnings for male lawyers are above those for female lawyers throughout 2000–2010.
- The gap does not disappear as cohorts age.

Economic message. The raw career-outcome gap is persistent, which motivates asking whether differences in performance help explain it.

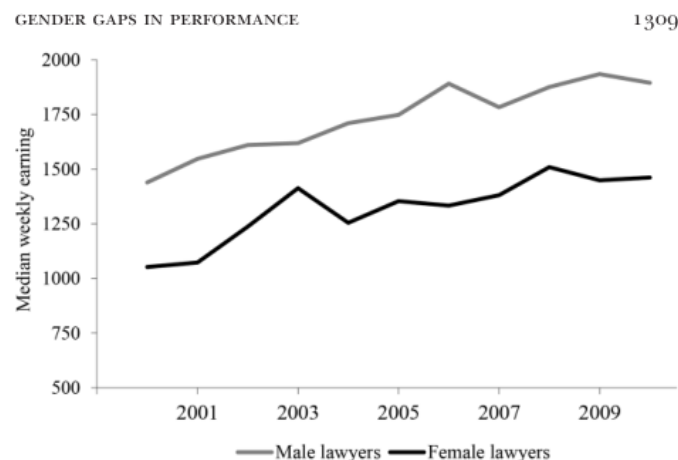


FIG. 1.—Evolution of lawyers' gender gap in earnings, 2000–2010. Median weekly earnings for lawyers in the period 2000–2010. Source: Current Population Survey.

are law firm partners. This divergence also appears to persist over time as

5.2 Table 1: descriptive statistics

TABLE 1
DESCRIPTIVE STATISTICS

	MALE LAWYERS			FEMALE LAWYERS			<i>p</i> -VALUE
	Observations	Mean	Standard Deviation	Observations	Mean	Standard Deviation	
Total earnings (\$)	684	150,667	74,531	441	132,685	70,282	.00
Hours billed (annual)	684	1,826	535	441	1,677	520	.00
New client revenue (\$)	684	53,346	171,965	441	23,349	68,892	.00
Target hours to bill	458	1,827	144	304	1,759	201	.01
Hours worked (per week)	684	54.09	12.80	441	48.83	13.84	.00
Age	684	36.12	4.98	441	35.29	4.92	.01
Marriage	684	.81	.39	441	.75	.43	.02
Children	684	1.22	1.24	441	.82	.91	.00
White	684	.83	.38	441	.75	.43	.00
Tenure (years)	684	5.18	2.49	441	5.26	2.44	.59
Private law firm	684	.92	.27	441	.93	.26	.57
Size of workplace > 100	684	.48	.50	441	.51	.50	.26
Law school ranking	597	4.95	1.08	392	5.05	1.10	.17
Undergraduate university ranking	662	12.89	3.50	435	13.04	3.62	.48
Judicial clerk	684	.02	.15	441	.03	.17	.44
Moot court	684	.32	.47	441	.35	.48	.31
General journal	684	.22	.42	441	.20	.40	.39
Specific journal	684	.20	.40	441	.25	.44	.05

NOTE.—Total earnings are calculated as a sum of straight salary and bonus (expressed in US dollars). Hours billed (annual) is the number of hours billed last year (2006). New client revenue is the approximate amount of new client revenue (expressed in US dollars) generated last year (2006). Target hours to bill is the total number of hours the lawyer was expected to bill in the previous year (2006) by the law firm for which the lawyer worked, conditional on having a strictly positive number of target hours. Marriage takes the value one if the lawyer is married, remarried after divorce, or in a domestic partnership and zero if single, divorced or separated, widowed, or other. Children refers to the lawyer's number of children. White takes the value one if the lawyer is Caucasian and zero if the lawyer is a member of a minority group (black, Hispanic, Native American, and Asian). Tenure is the number of years that the lawyer has been working for the current employer. Private law firm takes the value one if the lawyer works in a private law firm and zero if the lawyer works for another organization. Size of workplace > 100 takes the value one if the number of individuals employed in the organization is greater than 100 and zero otherwise. Hours worked (per week) is the number of hours spent working last week (at the office or away from the office). Undergraduate university ranking and law school ranking are bracketed rankings based on the *U.S. News and World Report* of 1996 and 2003 for undergraduate and law school studies, respectively. Both variables are redefined such that the higher the value, the more prestigious the educational institution. Judicial clerk takes the value one if the lawyer has held a position as a judicial clerk in state or federal courts and zero otherwise. Moot court takes the value one if the lawyer participated in simulated mock trials as a student and zero otherwise. General (specific) journal takes the value one if the lawyer participated in law journal editorial activities as a student and zero otherwise.

Read:

- Average earnings, hours billed, and new client revenue are all higher for men.
- The raw new-client-revenue gap is especially large.
- Women have fewer children on average and work fewer hours per week, but they are similar in tenure and law-firm employment.

Economic message. The performance gaps are not hidden inside regressions; they are immediately visible in the sample means.

5.3 Table 2 and Figure 2: baseline performance gaps and their distribution

TABLE 2
PERFORMANCE GAPS

	HOURS BILLED			NEW CLIENT REVENUE		
	(1)	(2)	(3)	(4)	(5)	(6)
Female	-.153*** [.0329]	-.103*** [.0315]	-.0971*** [.0319]	-.299*** [.0916]	-.293*** [.102]	-.324*** [.104]
Age		-.0132*** [.0032]	-.0116*** [.0033]		-.0091 [.0102]	-.0118 [.0109]
Married		.0595 [.0396]	.0645 [.0401]		.242* [.127]	.243* [.131]
1 child		-.0174 [.0510]	.0001 [.0520]		.0531 [.164]	.0927 [.170]
2 children		.0016 [.0501]	-.0169 [.0512]		-.0357 [.161]	-.074 [.168]
3+ children		-.0953 [.0625]	-.0785 [.0635]		-.107 [.201]	-.119 [.208]
Child aged < 4		-.0159 [.0454]	-.0164 [.0462]		-.147 [.146]	-.188 [.151]
White		-.0127 [.0377]	-.0277 [.0389]		.0319 [.122]	.0153 [.127]
Tenure		.0141** [.0059]	.0126** [.006]		.0388** [.0190]	.0413** [.0197]
Full-time		.500*** [.0623]	.498*** [.0629]		.165 [.200]	.126 [.206]
Undergraduate university ranking			-.00124 [.0042]			-.0113 [.0138]
Law school ranking			.0097 [.0156]			.0461 [.0511]
Judicial clerk			.114 [.0889]			.732** [.291]
Moot court			.0098 [.0301]			.0737 [.0986]
General journal			.0833** [.0353]			-.002 [.116]
Specific journal			.0761** [.0352]			-.0026 [.115]
Constant	1.842*** [.0205]	.683 [.478]	.584 [.487]	.527*** [.0571]	.168 [.1540]	.311 [.1593]
Firm controls	No	Yes	Yes	No	Yes	Yes
Region fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Observations	1,039	1,014	974	1,039	1,014	974
R ²	.021	.301	.311	.01	.066	.08

NOTE.—Hours billed is the annual number of hours billed (expressed in 1,000s of hours) last year (2006), and new client revenue is the approximate amount of new client revenue (expressed in 100,000s of US dollars) generated last year (2006). Firm controls include the type of organization (solo practice, private law firm, other Fortune 1,000 industry/service, other business/industry, labor union, trade association, other) and the size of the organization, which are bracketed (0–5, 6–10, 11–25, 25–50, 51–100, 101–150, 151–200, 201–250, 251–500, 501–1,000, and 1,000+). For definitions of other variables, see table 1.

GENDER GAPS IN PERFORMANCE

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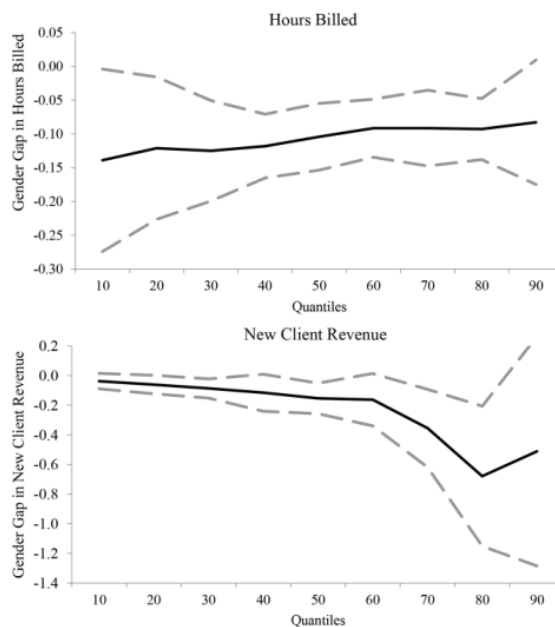


FIG. 2.—Percentiles of the gender gaps in performance. This figure provides the gender coefficient of the quantile regressions ranging from 0.10 to 0.90 for hours billed (expressed in 1,000s of hours) and new client revenue (expressed in 100,000s of US dollars) generated last year, after controlling for individual, education, and firm characteristics as

Figure 1: Baseline performance evidence from the paper.

Read:

- Table 2 shows that men bill about 153 more hours annually than women in the raw data.
- After adding individual, firm, and education controls, a gap of roughly 100 billed hours remains.
- Men also bring in more than twice as much new client revenue, and the revenue gap remains about \$30,000 after controls.
- Figure 2 shows that the hours-billed gap is fairly stable across the distribution, while the new-client-revenue gap is especially large in the upper tail.

Economic message. The paper’s central claim starts here: gender differences in measured performance are sizable and not explained away by standard observables.

5.4 Table 3 and Table 4: target hours and performance per hour worked

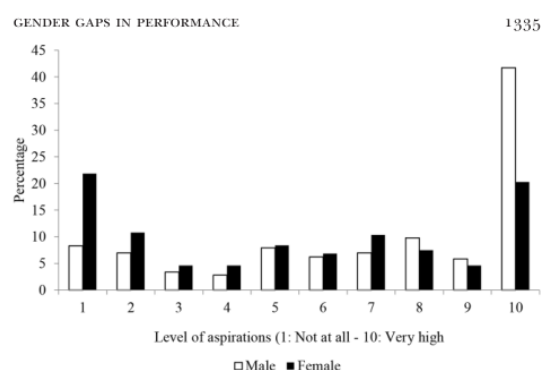


FIG. 3.—Aspirations to become partner. Percentage of responses by gender to the question “How strongly do you aspire to attain an Equity Partner position within your firm?” with possible answers ranging from 1 = not at all to 10 = very high (After the JD study, 2007).

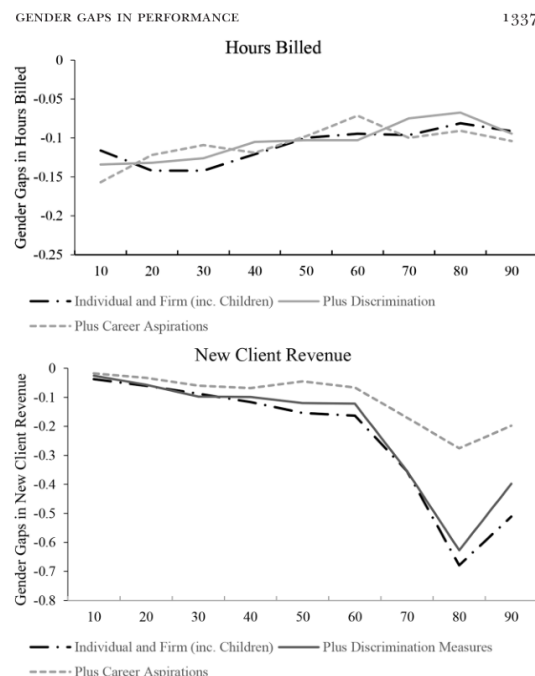


FIG. 4.—Percentiles of the gender gaps in performance with controls. This figure provides the gender coefficient of the quantile regressions ranging from 0.10 to 0.90 for hours

Read:

- Table 3 shows no statistically meaningful gender gap in target billable hours once controls are included.
- There is also no gender difference in the probability of reporting zero target hours.
- Table 4 shows no significant gender gap in billed-hours-per-hour-worked, but there is a significant gap in client-revenue-per-hour-worked and in aggregate performance per hour worked.

Economic message. Women are not simply being assigned systematically lower formal billing targets, and the performance gap is not reducible to hours worked alone.

5.5 Tables 6–8: discrimination channels

Read:

- The paper studies not-enough-assignments, partner discounting of hours, mentor characteristics, client-facing tasks, and self-reported discriminatory experiences.
- These variables matter little for explaining the gender coefficient.

Economic message. Directly measurable workplace discrimination channels do not account for much of the performance gap in this sample.

	CONSTRAINED	
	(1)	(2)
Female	.0001 [.0276]	.0011 [.0569]
1,600 < hours billed < 1,800	.0483 [.0402]	.0534 [.0551]
1,800 < hours billed < 2,100	-.122*** [.0388]	-.120** [.0509]
2,100 < hours billed < 3,000	-.245*** [.0442]	-.250*** [.0562]
Female $\times$ (1,600 < hours billed < 1,800)		-.0093 [.0778]
Female $\times$ (1,800 < hours billed < 2,100)		-.0068 [.0699]
Female $\times$ (2,100 < hours billed < 3,000)		.0182 [.0827]
Constant	.269 [.415]	.267 [.416]
Individual controls	Yes	Yes
Firm controls	Yes	Yes
Region fixed effects	Yes	Yes
Education controls	Yes	Yes
Observations	974	974
R ²	.122	.122

* Significant at the 10 percent level.
 ** Significant at the 5 percent level
 *** Significant at the 1 percent level.

	Hours Billed (1)	New Client Revenue (2)	Hours Billed (3)	New Client Revenue (4)	Hours Billed (5)	New Client Revenue (6)
Female	-.0992*** [.0325]	-.329*** [.106]	-.105*** [.0322]	-.368*** [.105]	.0881*** [.0329]	-.326*** [.105]
Senior mentor	.0066 [.0529]	.0141 [.173]				
Male mentor	.119** [.0549]	-.0386 [.193]				
Senior mentor × male mentor	.122 [.0748]	.0065 [.245]				
Taxi						
Client meeting			.004 [.012]	.108*** [.0391]		
Keep client updated			.012 [.0105]	.00175 [.0342]		
Formulate strategy			.02 [.01]	.02 [.0327]		
Traveling to court/clients			.0164 [.0112]	-.0291 [.0364]		

NOTE.—Senior mentor refers to whether the lawyer's mentor is a law firm partner. Male mentor refers to whether the lawyer's mentor is male. Perceived discrimination refers to questions regarding experiencing demeaning comments, missing out on a desirable assignment, client requests that someone else handle a matter, having a colleague or supervisor requesting someone else to handle a matter, or other form of discrimination. Task refers to tasks carried out by the lawyer responsible for keeping the client updated, being involved in formulating strategy, traveling to make court appearances or to meet clients, or holding face-to-face meetings with clients. The summary statistics for these variables can be found in online table A.2. For definitions of other variables, see table 1.

* Significant at the 10 percent level.
 ** Significant at the 5 percent level.
 *** Significant at the 1 percent level.

5.6 Table 9, Figure 3, Figure 4, and Tables 10–13: children, aspirations, and alternative channels

	HOURS BILLED		NEW CLIENT REVENUE	
	(1)	(2)	(3)	(4)
Female	-.0971***	-.0569	-.524***	-.542**
	[.0310]	[.0480]	[.1801]	[.1480]
1 child	.0001	-.0305	.0927	.184
	[.0020]	[.0681]	[.1707]	[.2241]
2 children	-.0109	-.0633	-.074	-.0436
	[.0512]	[.0625]	[.1168]	[.205]
3+ children	-.0785	-.0996	-.119	-.141
	[.0635]	[.0751]	[.208]	[.249]
Child aged < 4	-.0164	.009	-.188	-.255
	[.0462]	[.0558]	[.151]	[.183]
Female × 1 child		.0945	.221	.221
		[.101]	[.531]	[.531]
Female × 2 children		.142	-.088	-.088
		[.0995]	[.327]	[.327]
Female × 3+ children		.0022	.249	.249
		[.140]	[.459]	[.459]
Female × child aged < 4		-.256***	.197	.197
		[.0885]	[.294]	[.294]
Constant	.584	.635	.511	.518
	[.467]	[.497]	[1.593]	[1.600]
Individual controls	Yes	Yes	Yes	Yes
Firm controls	Yes	Yes	Yes	Yes
Region fixed effects	Yes	Yes	Yes	Yes
Education controls	Yes	Yes	Yes	Yes
Observations	974	974	974	974
R ²	.311	.32	.08	.081

NOTE:—Children aged < 4 takes the value one if the lawyer has a child 3 years of age or younger. For definitions of the other variables, see table 1.

* Significant at the 10 percent level.

** Significant at the 5 percent level.

*** Significant at the 1 percent level.

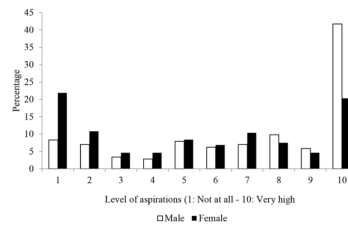


FIG. 3.—Aspirations to become partner. Percentage of responses by gender to the question “How strongly do you aspire to attain an Equity Partner position within your firm?” with possible answers ranging from 1 = not at all to 10 = very high (After the JD study, 2007).

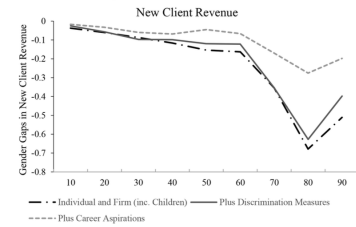
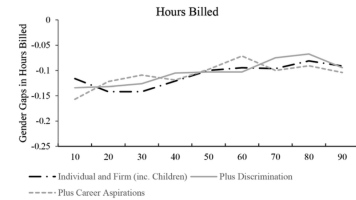


FIG. 4.—Percentiles of the gender gaps in performance with controls. This figure provides the gender coefficient of the quantile regressions ranging from 0.10 to 0.90 for hours billed (expressed in 1,000s of hours) and new client revenue (expressed in 100,000s of US dollars). The gaps presented are for (i) after controlling for individual, education, and firm characteristics and region fixed effects; (ii) controlling for all factors in part i as well as measures of discrimination; and (iii) controlling for all factors in part i as well as for long-term aspirations.

Read:

- Table 9 shows that women with children under age 4 bill substantially fewer hours; the comparable effect for men is absent.
- Figure 3 shows a large aspiration gap: 60% of men report aspirations of 8 or above to become partner, compared with 32% of women.
- Table 10 shows that aspirations strongly predict both billed hours and new client revenue, and they largely explain the remaining new-client-revenue gap.
- Figure 4 shows that controlling for aspirations closes much of the new-client-revenue gap throughout the distribution.
- Table 12 shows that predicted aspirations based on early variables produce a similar message.
- Table 13 shows that overbilling, networking, and weekend work influence performance levels but explain little of the gender gap.

Economic message. The core mechanism section of the paper says: young children help explain the billed-hours gap; aspirations help explain the client-revenue gap.

5.7 Tables 14–16 and Figure 5: earnings and the role of performance

Read:

- Without performance measures, the raw gender gap in log annual earnings is 18 log points, and standard controls explain about half of it.
- Once billed hours and new client revenue are added, nearly half of the remaining gap is explained.
- Billing 100 additional hours is associated with about a 3.2% increase in salary.

	HOURS BILLED		NEW CLIENT REVENUE	
	(1)	(2)	(3)	(4)
Female	-.0848** [.0344]	-.133* [.0722]	-.146 [.139]	-.0803 [.272]
Aspirations	.0224*** [.0050]	.0188*** [.0063]	.0662*** [.0190]	.0711*** [.0260]
Female × aspirations		.0075 [.0008]		-.0105 [.0371]
Constant	.509 [.500]	-.927 [.501]	-.568 [1.883]	
Individual controls	Yes	Yes	Yes	Yes
Firm controls	Yes	Yes	Yes	Yes
Education controls	Yes	Yes	Yes	Yes
Region fixed effects	Yes	Yes	Yes	Yes
Observations	617	617	617	617
R ²	.31	.311	.075	.075

NOTE.—Hours billed is expressed in 1,000s of hours. New client revenue is expressed in 100,000s of US dollars. Aspirations refers to how strongly the lawyer aspires to obtain an equity partnership. The variable takes values from 1 to 10, where 1 represents not at all and 10 represents very high. For definitions of other variables, see table 1.
* Significant at the 10 percent level.
** Significant at the 5 percent level.
*** Significant at the 1 percent level.

	Hours Billed (1)	New Client Revenue (2)	Hours Billed (3)	New Client Revenue (4)	Hours Billed (5)	New Client Revenue (6)
Female	-.121*** [.0495]	-.289** [.1182]	-.161*** [.0461]	-.146 [.120]	-.262*** [.0525]	-.1161 [.1064]
Aspirations					-.0029** [.0010]	-.0119** [.0048]
Not enough engagements	-.1113*** [.0080]	-.473 [.141]			-.0099* [.0051]	1.111 [.411]
Partner discussed hours					-.0031 [.0010]	-.0037 [.0011]
Partner discussed tenure					-.008 [.0030]	-.1027 [.0190]
Senior mentor					-.0026 [.0010]	-.0118 [.0190]
Male mentor	-.0013 [.0010]	-.0119 [.0010]			-.0005 [.0006]	-.0020 [.0011]
Senior mentor × male mentor					-.0002 [.0006]	-.0001 [.0006]
Deceived discrimination					.12 [.0096]	-.0707 [.141]
Deceiving comments	-.0029* [.0010]	-.0084 [.0010]			-.0041 [.0010]	-.005 [.0011]
Mixed deceptive engagement					-.0011 [.0010]	-.0011 [.0011]
Client request another	.0013 [.0010]	-.0146 [.0010]			-.0011 [.0010]	-.0011 [.0011]
Superior request another	.0013 [.0010]	-.0022 [.0010]			-.0011 [.0010]	-.0011 [.0011]
Other					-.0011 [.0010]	-.0011 [.0011]

Task						
Client meeting	-.012 [.0071]	.1119* [.0495]			-.08 [.0071]	.0007 [.0001]
Keep client updated	.014 [.0010]	.1441** [.0010]			.012 [.0010]	.1301** [.0010]
Particular strategy	.002 [.0010]	-.016 [.0010]			.002 [.0010]	.001 [.0010]
Traveling to court/client	.014 [.0010]	-.014 [.0010]			.014 [.0010]	.014 [.0010]
					.007 [.0010]	.1119** [.0010]
					.007 [.0010]	.1119** [.0010]
Individual controls	Yes	Yes	Yes	Yes	Yes	Yes
Firm controls	Yes	Yes	Yes	Yes	Yes	Yes
Education controls	Yes	Yes	Yes	Yes	Yes	Yes
Region fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Observations	617	617	617	617	617	617

NOTE.—Hours billed is expressed in 1,000s of hours. New client revenue is expressed in 100,000s of US dollars. See tables 1, 6, 8, and 10 for definitions of all variables.
* Significant at the 10 percent level.
** Significant at the 5 percent level.
*** Significant at the 1 percent level.

	HOURS BILLED		NEW CLIENT REVENUE		HOURS BILLED		NEW CLIENT REVENUE	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Female	-.0848** [.0344]	-.0847** [.0321]	-.146 [.139]	-.199 [.189]	-.0574 [.0438]	-.0541 [.0454]	-.0188 [.165]	-.0457 [.178]
Aspirations	.0224*** [.00504]	.0218*** [.00537]	.0662*** [.0190]	.0202** [.0212]				
Predicted aspirations					.0359** [.0162]	.0396** [.0182]	.149** [.0603]	.161** [.0713]
Constant	.509 [.500]	1.713*** [.427]	-.927 [.501]	-1.114 [1.883]	.392 [.507]	.539 [.511]	-1.819 [1.940]	-2.247 [2.047]
Individual controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Firm controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Education controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Region fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Controls from table 11	No	No	No	No	No	No	No	No
Observations	617	617	617	617	587	587	587	587

NOTE.—Hours billed is expressed in 1,000s of hours. New client revenue is expressed in 100,000s of US dollars. Aspirations refers to how strongly the lawyer aspires to obtain an equity partnership. The variable takes values from 1 to 10, where 1 represents not at all and 10 represents very high. Predicted aspirations refers to aspirations as predicted by the following questions: (1) How satisfied are you with your decision to become a lawyer? (2) How much longer would you like to stay with your current employer? Both questions were asked in the first wave (2002). See tables 1, 6, 8, and 10 for definitions of all variables.
* Significant at the 10 percent level.
** Significant at the 5 percent level.
*** Significant at the 1 percent level.

	HOURS BILLED		NEW CLIENT REVENUE		HOURS BILLED		NEW CLIENT REVENUE		HOURS BILLED		NEW CLIENT REVENUE	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
Female	-.0808*** [.0320]	-.0900*** [.0348]	-.315*** [.105]	-.337*** [.114]	-.0005*** [.0342]	-.127*** [.0392]	-.370*** [.110]	-.326*** [.126]	-.0030*** [.0326]	-.130*** [.0400]	-.312*** [.109]	-.270*** [.133]
Less than others	-.103*** [.0405]	-.128** [.0552]	-.128 [.133]	-.189 [.181]								
Female × less than others												
Network time					.0017 [.0039]	-.0025 [.0043]	.0310** [.0126]	.0570** [.0143]				
Female × network									.0143*** [.00370]	.0108** [.0043]	.0200** [.0125]	.0300** [.0144]
Weekend time												
Female × weekend												
Constant	.587 [.486]	.59 [.486]	.316 [.486]	.322 [.486]	1.350*** [1.594]	1.360*** [.495]	.611 [.494]	.589 [.591]	.575 [.490]	.575 [.490]	.412 [.633]	.411 [1.634]
Individual controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Firm controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Education controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Region fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	974	974	974	974	903	903	903	903	930	930	930	930

NOTE.—Hours billed is expressed in 1,000s of hours. New client revenue is expressed in 100,000s of US dollars. Less than others takes the value one if the lawyer responds that he or she is less likely to bill for actual hours worked compared to colleagues as a reason for the difficulty in meeting billable and zero otherwise. Net work time is the amount of time a lawyer spends attending networking functions and/or participating in recreational activities (e.g., golf) for networking purposes with other lawyers or clients. Weekend time is the amount of time a lawyer spends working away from the office or firm on weekends.
* Significant at the 10 percent level.
** Significant at the 5 percent level.
*** Significant at the 1 percent level.

	LOG(ANNUAL EARNINGS)				
	(1)	(2)	(3)	(4)	(5)
Female	-.181*** [.0343]	-.0997*** [.0359]	-.119*** [.0342]	-.100*** [.0307]	-.100*** [.0313]
Age		-.0151*** [.00351]	-.0136*** [.0035]	-.0041 [.0031]	-.0023 [.0033]
Married		.0262 [.0452]	.0189 [.0445]	-.0076 [.0386]	.0042 [.0393]
1 child		-.0047 [.0583]	-.0281 [.0574]	.0067 [.0497]	.0251 [.0509]
2 children		.0034 [.0570]	-.0197 [.0559]	.039 [.0488]	.0302 [.0502]
3+ children		.107 [.0712]	.129* [.0701]	.0825 [.0609]	.0961 [.0622]
Child aged < 4		.0121 [.0517]	7.88E-03 [.0508]	.0019 [.0443]	-.017 [.0453]
White		.0597 [.0427]	.0808* [.0421]	.0297 [.0368]	.0313 [.0381]
Tenure		.0414*** [.0066]	.0385*** [.0065]	.0195*** [.0057]	.0175*** [.0059]
Full-time		.519*** [.0697]		.488*** [.0607]	.504*** [.0616]
Hours worked			.228*** [.0244]		
Constant	11.81*** [.0214]	11.53*** [.156]	11.39*** [.154]	11.59*** [.466]	11.31*** [.477]
Region fixed effects	Yes	Yes	Yes	Yes	Yes
Firm controls	No	No	No	Yes	Yes
Education controls	No	No	No	No	Yes
Observations	1,039	1,014	1,014	1,014	974
R ²	.027	.131	.158	.388	.403

NOTE.—Hours worked is expressed in 1,000s of annual hours worked. For definitions of the variables, see table 1.
* Significant at the 10 percent level.
** Significant at the 5 percent level.
*** Significant at the 1 percent level.

	Annual Earnings
Female	-.29.48** [13.41]
Hours billed	26.482*** [4.851]
Female × hours billed	10.90 [7.12]
New client revenue	5.67*** [1.254]
Female × new client revenue	5.025 [4.186]
Constant	27.82 [28.49]
Individual controls	Yes
Firm controls	Yes
Region fixed effects	Yes
Education controls	Yes
Observations	974
R ²	.424

* Significant at the 10 percent level.
** Significant at the 5 percent level.
*** Significant at the 1 percent level.

- Generating an additional \$100,000 in new client revenue is associated with about a 4.0% increase in salary.
- Figure 5 shows that the earnings gap persists across the distribution, especially in the upper tail.
- Table 16 shows no meaningful gender difference in the pay return to each billed hour or each dollar of client revenue.

Economic message. The paper’s main payoff is here: once actual performance is observed, a large portion of the “unexplained” gender earnings gap is no longer unexplained.

FIGURES IN TABLE 15, WE INCLUDE THE MAIN PERFORMANCE VARIABLES, HOURS

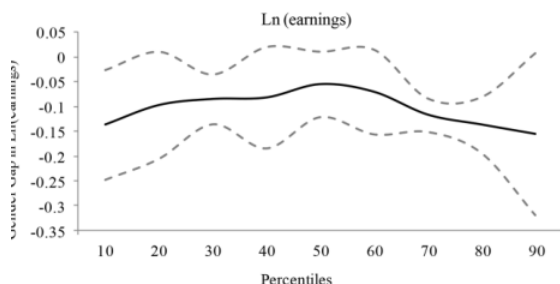


FIG. 5.—Percentiles of the gender gaps in earnings. This figure provides the gender coefficient of the quantile regressions ranging from 0.10 to 0.90 for $\ln(\text{earnings})$, after controlling for individual, education, and firm characteristics, as well as region fixed effects. confidence intervals (dashed line in gray) are at the 5 percent confidence level.

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Female	-0.10*** [0.031]	-0.081* [0.027]	-0.028* [0.009]	-0.022** [0.007]	-0.0245 [0.0202]	-0.0295 [0.0202]	-0.0273 [0.0203]
Hours worked			0.0022*** [0.0002]	0.0033*** [0.0001]	0.004*** [0.0001]	0.0040*** [0.0001]	0.0041*** [0.0001]
Hours billed			0.0022*** [0.0001]	0.0021*** [0.0001]	0.0021*** [0.0001]	0.0021*** [0.0001]	0.0021*** [0.0001]
New client revenue			0.0001*** [0.0001]	0.0001*** [0.0001]	0.0001*** [0.0001]	0.0001*** [0.0001]	0.0001*** [0.0001]
Hours billed (squared)							
New client revenue (squared)							
Hours billed (cubic)							
New client revenue (cubic)							
Constant	11.31*** [4.77]	11.12*** [4.40]	10.90*** [4.47]	10.21*** [4.19]	9.179*** [4.37]	8.950*** [4.42]	11.13*** [4.27]
Individual controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Full-time control	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Firm controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Region fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Education controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Areas of law	No	No	No	No	No	No	No
Observations	974	974	974	974	974	974	974
R ²	.403	.472	.482	.44	.544	.55	.55

NOTE.—Hours billed is expressed in 1,000s of hours. New client revenue is expressed in 100,000s of US dollars. Hours worked is expressed in 1,000s of annual hours worked. For definitions of the variables, see Table 1.
* Significant at the 10 percent level.
** Significant at the 5 percent level.
*** Significant at the 1 percent level.

5.8 Table 17: promotion to partner

Read:

- The raw partnership gap is close to 10 percentage points.
- Hours worked explain little of that gap.
- Earlier billed hours and new client revenue strongly predict later partnership.
- Once performance is accounted for, the female coefficient becomes statistically insignificant.
- Early aspirations are also positively related to making partner.

Economic message. Performance matters not only for current pay but also for the dynamic career margin that drives future pay: promotion.

percent of the gap remaining after controlling for individual and

TABLE 17
PARTNERSHIP

	PARTNERSHIP					
	(1)	(2)	(3)	(4)	(5)	(6)
	-.0913**	-.105**	-.0948**	-.0596	-.0581	-.012
	[.0393]	[.0468]	[.0473]	[.0488]	[.0491]	[.0630]
worked			.0733*		.0143	.0361
			[.0408]		[.0430]	[.0582]
billed				.239***	.235***	.251***
				[.0563]	[.0580]	[.0818]
ent revenue				.0772***	.0762***	.0458
				[.0241]	[.0243]	[.0288]
ions						.0553***
						[.0100]
fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
ual controls	No	Yes	Yes	Yes	Yes	Yes
ontrols	No	Yes	Yes	Yes	Yes	Yes
ion controls	No	Yes	Yes	Yes	Yes	Yes
ations	689	655	655	655	655	454

Partnership takes the value one if the lawyer is an equity partner or shareholder in a law firm in which he or she works in 2012 and zero otherwise, using responses from the third wave of the AJD data. Hours billed is expressed in 1,000s of hours. New client revenue is expressed in 100,000s of US dollars. Hours worked is expressed in 1,000s of annual hours worked. For definitions of the variables, see table 1.

*Significant at the 10 percent level.

**Significant at the 5 percent level.

***Significant at the 1 percent level.

6 Short summary

- The paper studies gender gaps in workplace performance among young U.S. lawyers.
- Using direct profession-standard performance measures, it finds that men bill about 10% more hours and bring in far more new client revenue.
- The billed-hours gap is not explained by target hours, obvious firm-expectation differences, or directly measured discrimination channels.
- Young children help explain women's lower billed hours.
- Aspirations to become partner are strongly related to performance and explain much of the new-client-revenue gap.
- Performance explains a sizable portion of the gender gap in annual earnings and later partnership.
- The broader message is that performance is an essential omitted variable in the study of gender gaps in elite careers.

7 Conclusion

7.1 Core conclusion (what the paper establishes)

The paper establishes that gender gaps in career outcomes among high-skilled professionals are partly gender gaps in measured workplace performance. In law firms, male lawyers bill more hours and generate more new client revenue than female lawyers, and these performance differences account for a meaningful share of the observed gaps in earnings and promotion.

7.2 Economic intuition

Two ideas stand out.

First, **hours worked is not performance**. A lawyer can work many hours without converting those hours into billable revenue or new clients. The profession rewards the latter.

Second, **different constraints and different career concerns produce different performance paths**. The presence of very young children appears to reduce women's billed hours much more than men's, while aspirations to become partner are especially important for originating new client business.

7.3 Takeaway

The paper's main contribution is not merely that women earn less or are promoted less. It is that once we observe the profession's own performance measures, a large part of the remaining gap in pay and promotion becomes understandable. That makes this paper especially important for thinking about inequality in performance-pay occupations.